Deep Learning Multimodal Report — SEED-IV

Generated: 20260423_131614

EEG Root: C:\Users\Rose J Thachil\Desktop\8th_sem_project\raw_data_project\dataset\eeg_raw_data

Eye Root: C:\Users\Rose J Thachil\Desktop\8th_sem_project\raw_data_project\dataset\eye_raw_data

Cross-Validation: 5-Fold GroupKFold (subject-based)

Model: EEG 1D-CNN + Eye Dense → Fusion → Softmax (4 classes)

Summary Metrics

Metric	Value
Mean Accuracy	0.2794
Std Accuracy	0.0034
Mean F1	0.1632
Std F1	0.0080
Folds	5

Per-Fold Metrics

Fold	Accuracy	F1 (Weighted)	Test Subjects
1	0.2758	0.1679	[5, 10, 15]
2	0.2808	0.1534	[4, 9, 14]
3	0.2786	0.1728	[3, 8, 13]
4	0.2772	0.1566	[2, 7, 12]
5	0.2844	0.1653	[1, 6, 11]

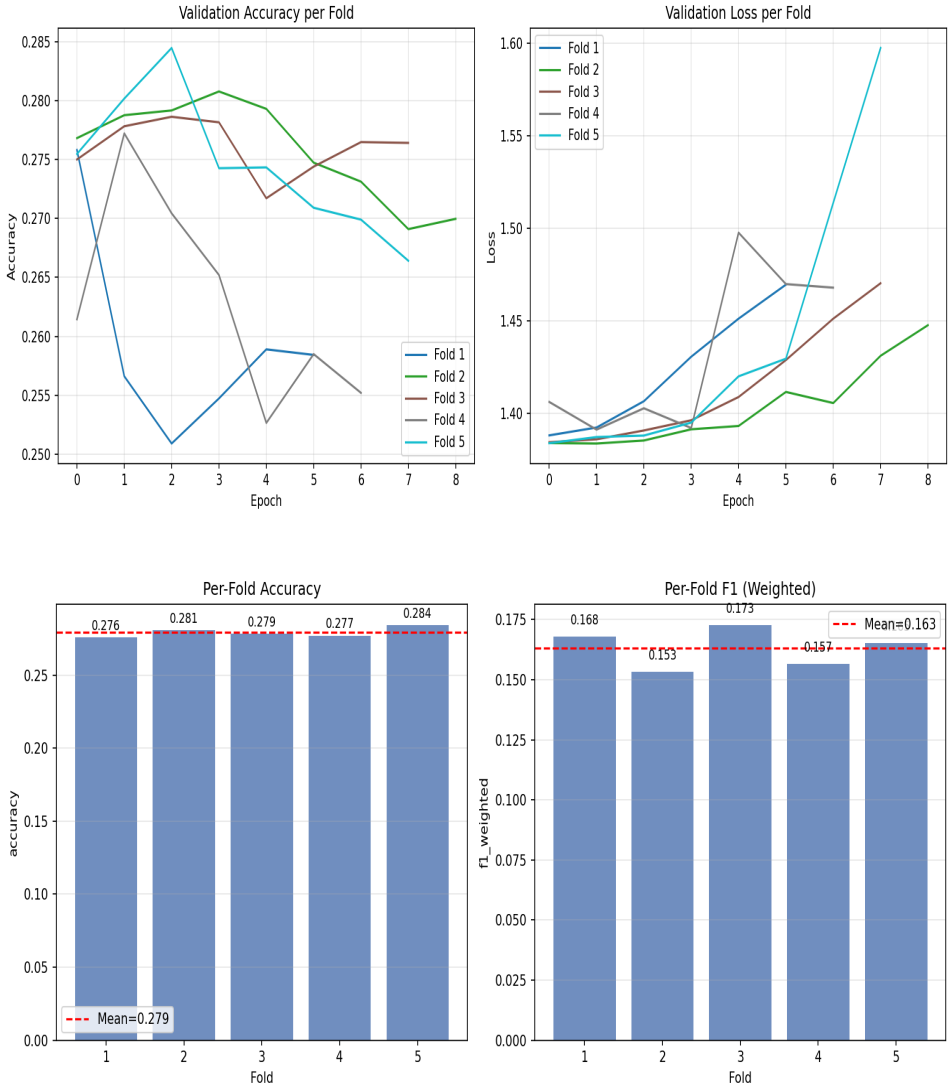
Model Architecture

EEG Branch: Input(time, 62 channels) → Conv1D(64,k7) → BN → Pool → Conv1D(128,k5) → BN → Pool → Conv1D(256,k3) → GlobalAvgPool → Dense(128) → Dropout(0.4) → Dense(64)

Eye Branch: Input(5 features) → Dense(64) → BN → Dropout(0.3) → Dense(32)

Fusion: Concatenate → Dense(128) → Dropout(0.4) → Dense(64) → Softmax(4)

Figures



Configuration

eeg_root: C:\Users\Rose J Thachil\Desktop\8th_sem_project\raw_data_project\dataset\eeg_raw_data

eye_root: C:\Users\Rose J Thachil\Desktop\8th_sem_project\raw_data_project\dataset\eye_raw_data

output_dir: C:\Users\Rose J Thachil\Desktop\8th_sem_project\raw_data_project\obj2

sfreq: 200

epoch_len_sec: 4

epoch_overlap: 0.5

n_eeg_channels: 62

eye_feature_dim: 5

n_classes: 4

n_folds: 5

epochs: 30

batch_size: 32

learning_rate: 0.001
random_state: 42
timestamp: 20260423_131614